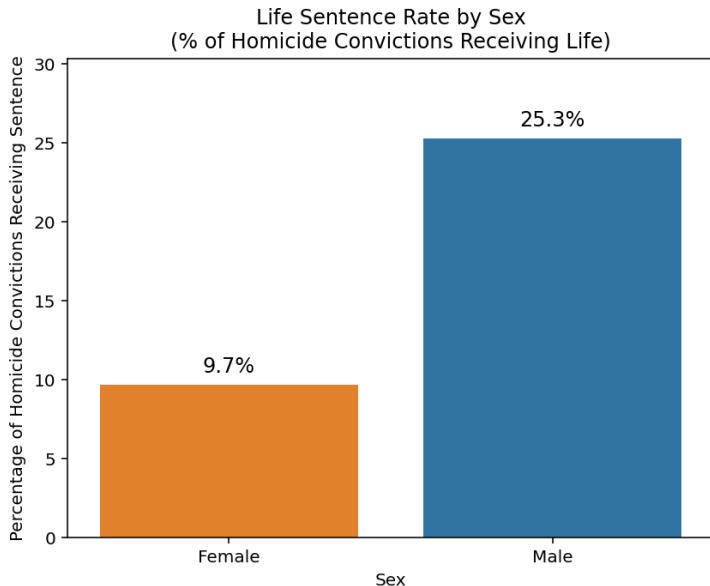
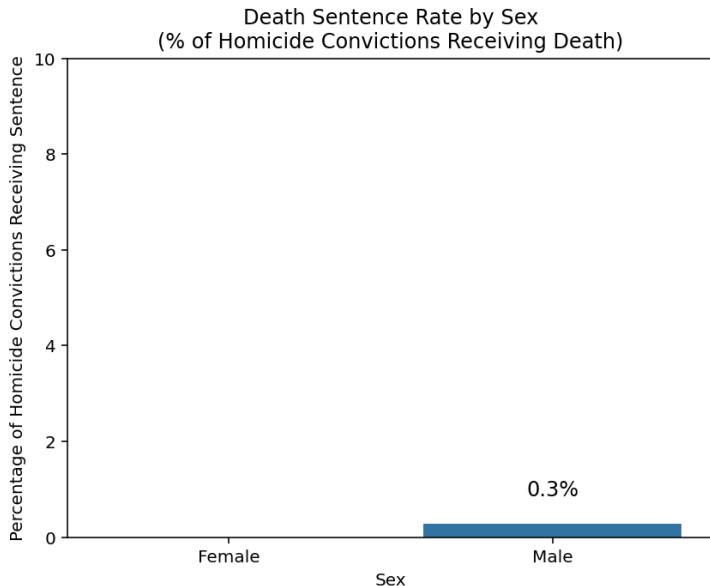


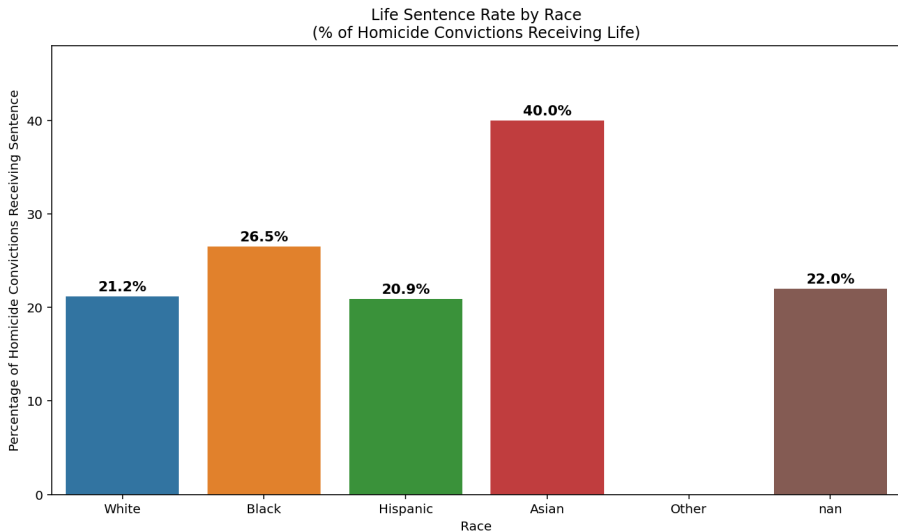
Life Sentence Rate By Sex



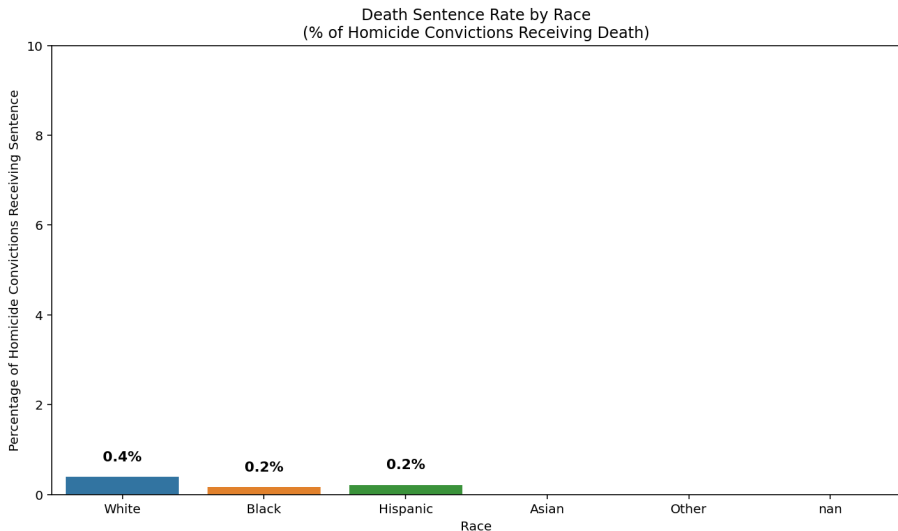
Death Sentence Rate By Sex



Life Sentence Rate By Race

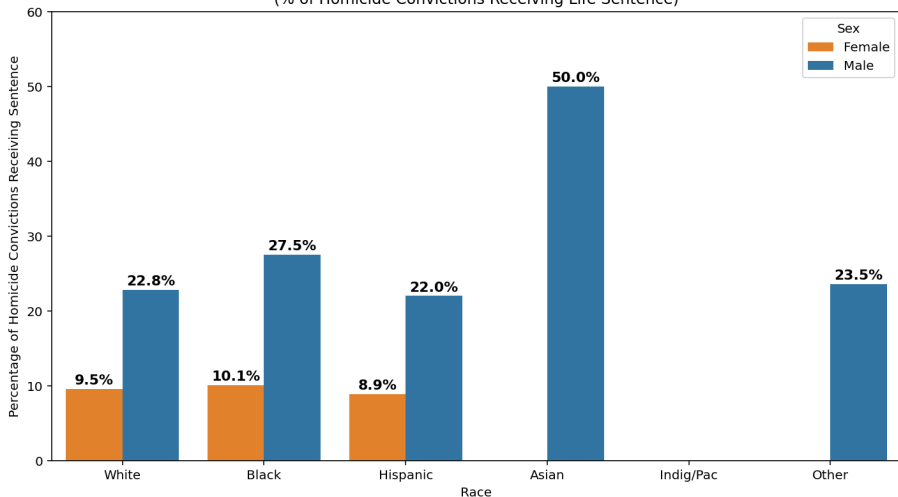


Death Sentence Rate By Race



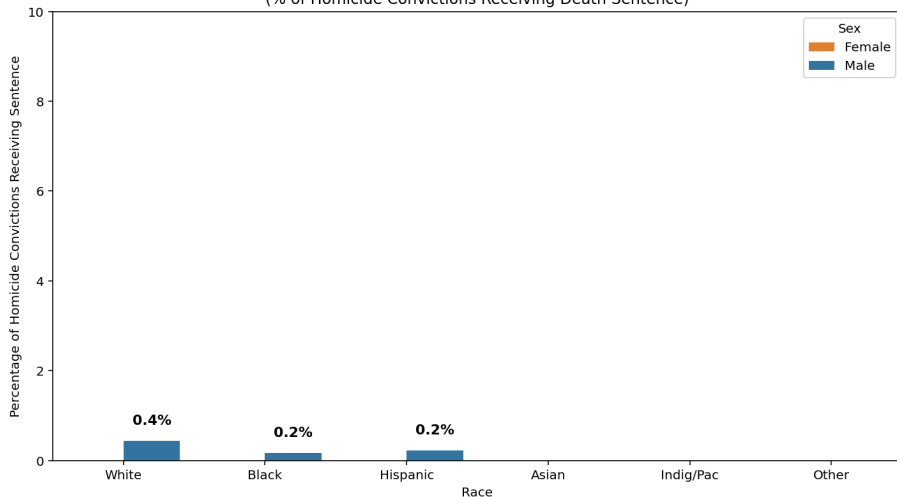
Life Sentence Rate By Sex and Race

Life Sentence Rate by Sex and Race
(% of Homicide Convictions Receiving Life Sentence)

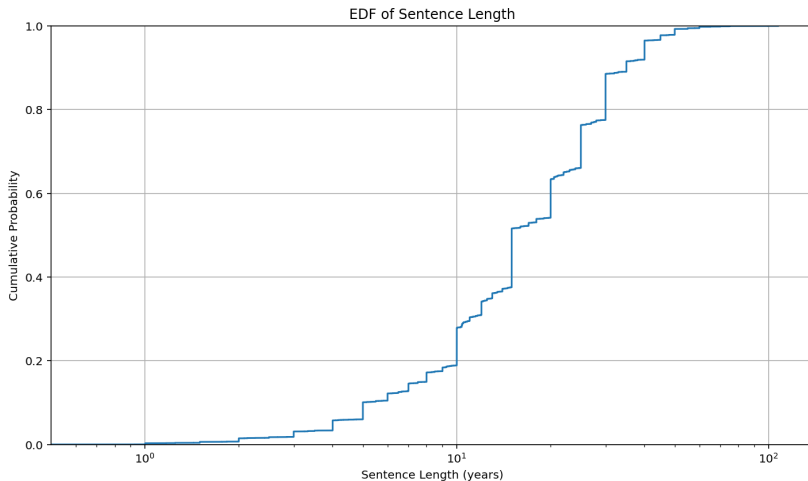


Death Sentence Rate By Sex and Race

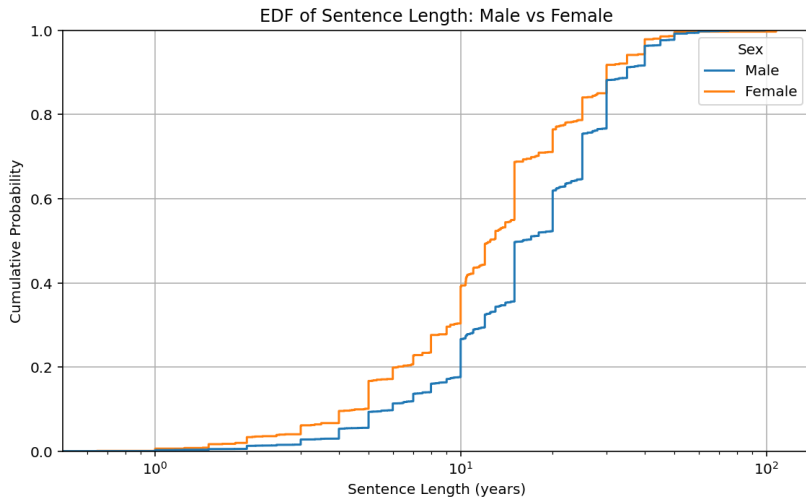
Death Sentence Rate by Sex and Race
(% of Homicide Convictions Receiving Death Sentence)



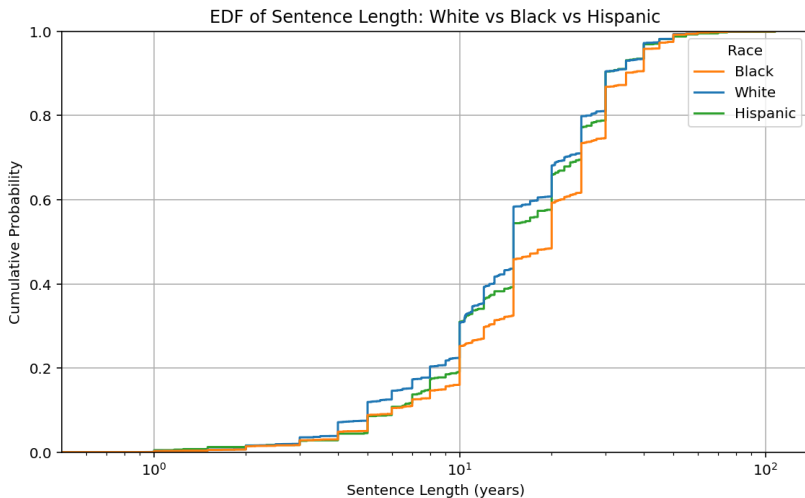
Empirical Distribution Function for All



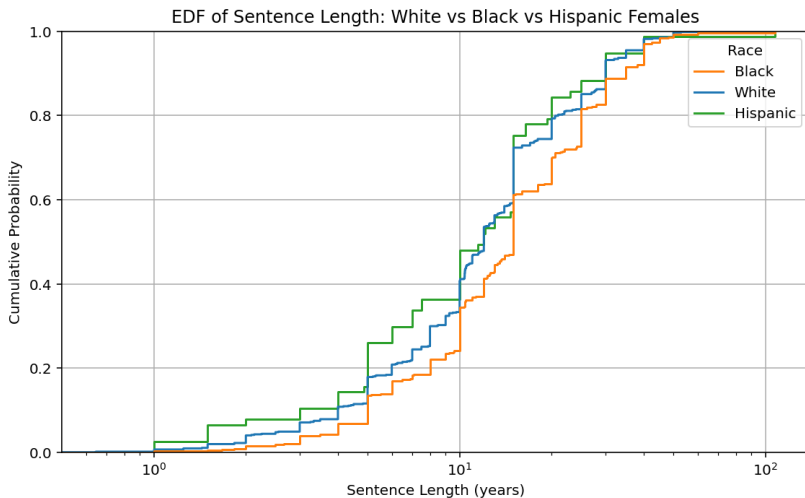
Empirical Distribution Function: Male vs Female



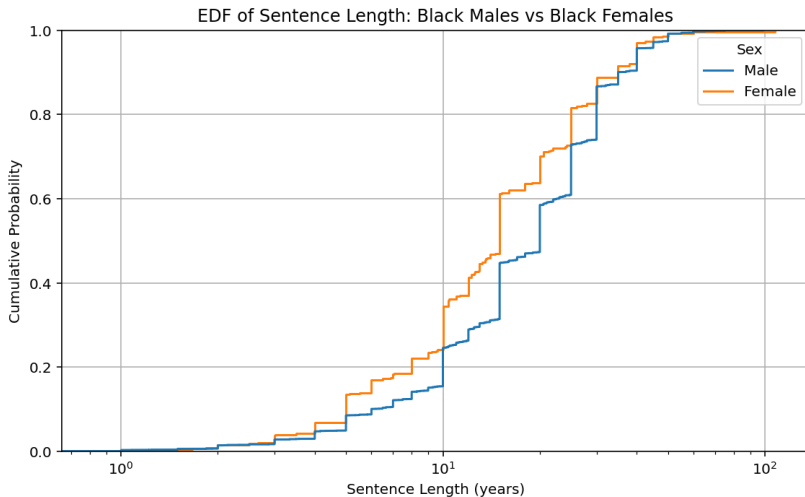
Empirical Distribution Function: Black vs White vs Hispanic



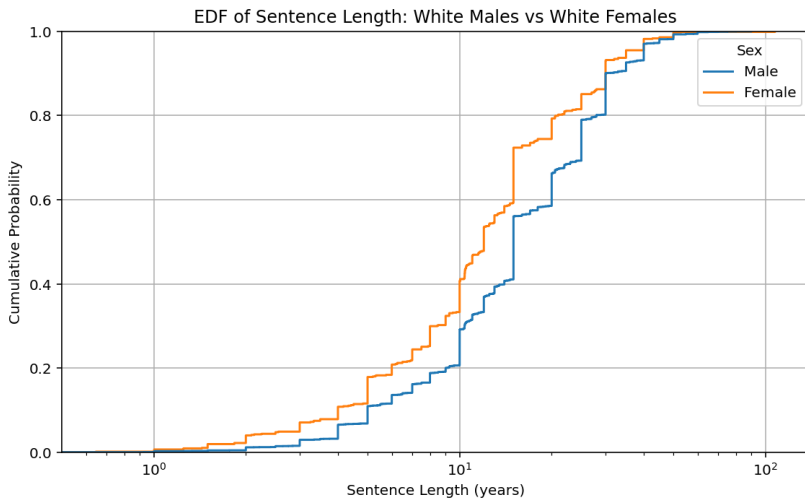
Empirical Distribution Function: Black Vs White Vs Hispanic Females



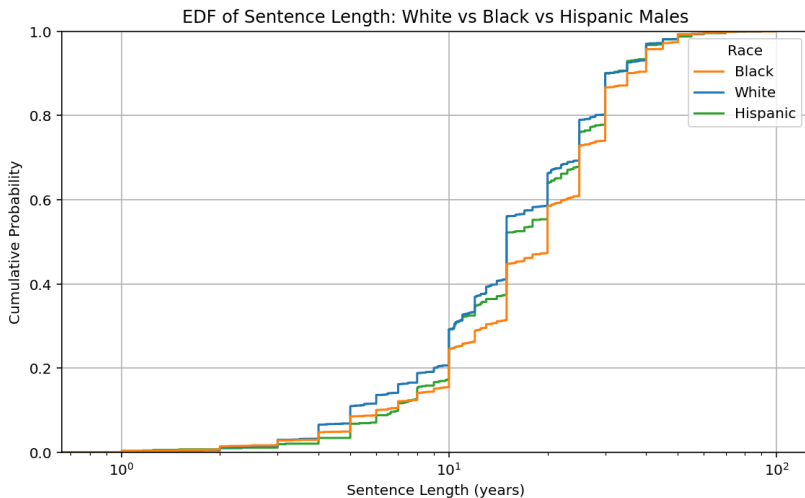
Empirical Distribution Function: Black Males vs Black Females



Empirical Distribution Function: White Males vs White Females



Empirical Distribution Function: Black Vs White Vs Hispanic Males



Lorenz Curve vs Generalized Lorenz Curve

Lorenz Curve Formula:

$$L\left(\frac{i}{N}\right) = \frac{\sum_{j=1}^i y_{(j)}}{\sum_{j=1}^N y_{(j)}}$$

Where:

- $y_{(j)}$ is the j -th smallest sentence length (sorted data).
- N is the total number of individuals.
- $i \in [1, N]$ is the population index.

Limitation:

- Comparing Lorenz curves requires groups (race or sex) to have the same mean sentence lengths μ .

Why Use the Generalized Lorenz Curve?

Generalized Lorenz Curve:

$$GLC\left(\frac{i}{N}\right) = \frac{1}{N} \sum_{j=1}^i y_{(j)} = \mu \cdot L\left(\frac{i}{N}\right)$$

Where:

- $y_{(j)}$ = j -th smallest sentence length.
- N = total number of individuals.
- μ is the mean sentence length.
- $L\left(\frac{i}{N}\right)$ = Lorenz Curve

In Our Context:

- Sentence lengths vary widely across race and sex.
- GLC allows us to capture differences in both distribution shape and total punishment.
- A higher GLC curve means more cumulative punishment burden for that group.

